

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 611 Operating Systems — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
MIT 6.1810 / 6.S081 Operating System Engineering — xv6 (free, with labs)	Primary source. A complete small Unix kernel with a book, lecture notes, and graded labs. The single best free OS course in existence.
MIT 6.S081 2020 archive — lecture video and labs	The version with full recorded lectures. Use alongside the current course page.
OSTEP — Operating Systems: Three Easy Pieces (free book)	The best written OS textbook, free from the authors. Organised as virtualisation, concurrency, persistence — which is this course's structure.
xv6: a simple, Unix-like teaching operating system (free book)	The commentary on the xv6 source. Read it with the code open beside it.
Berkeley CS162 (free lectures)	A third voice, with more emphasis on distributed systems and security.
Linux kernel documentation	What a production kernel actually does, for when you want to know how the real thing differs.

Module-by-module map

Module 01 — What an Operating System Is For

Resource	What it covers for this module
MIT 6.1810 — Lecture 1 (Introduction and Examples) and the xv6 book, Chapter 1	Sets up the whole course. Do the setup lab in the same sitting.
OSTEP — Chapters 2 and 4 (Introduction, The Abstraction: The Process)	The abstraction/arbitration framing, written unusually well.
xv6 book — Chapter 4 (Traps and system calls)	The system call path through actual code. Read it with trampoline.S open beside it.
RISC-V Privileged Architecture specification	The hardware contract for privilege modes and traps. Short, clear, and far more readable than the x86 equivalent.

Module 02 — Processes, Threads, and Context Switching

Resource	What it covers for this module
MIT 6.1810 — lectures and lab on scheduling / swtch	The context switch traced through real code, with the two-switch design explained.
xv6 book — Chapter 7 (Scheduling)	swtch, the scheduler loop, and the per-CPU scheduler context.
OSTEP — Chapters 4–6 (Processes, the Process API, Limited Direct Execution)	The clearest written treatment of fork/exec and of why the API is shaped as it is.
Baumann, Appavoo, Krieger, Roscoe — A fork() in the road (free)	A serious argument that fork was a mistake. Read it after you have understood why it was a good idea — the disagreement is instructive.

Module 03 — CPU Scheduling

Resource	What it covers for this module
OSTEP — Chapters 7–10 (Scheduling, MLFQ, Lottery, Multiprocessor)	The best free treatment of scheduling policy, and the source of the MLFQ rule formulation used here.
MIT 6.1810 — scheduling lecture and lab	xv6's scheduler in code, and the lab that replaces it.
Jones — What really happened on Mars (free)	The Pathfinder priority inversion, told by someone close to it. Short and worth reading in full.
Linux kernel documentation — CFS and EEVDF	How a production scheduler is actually structured.

Module 04 — Synchronization

Resource	What it covers for this module
OSTEP — Chapters 26–31 (Concurrency, Locks, Condition Variables, Semaphores)	The clearest written treatment of locks and condition variables, including the lost-wakeup derivation.

Resource	What it covers for this module
xv6 book — Chapter 6 (Locking)	Real spinlocks, including the interrupt and memory-barrier reasoning.
Paul McKenney — Is Parallel Programming Hard? (free book)	The practitioner's reference, and the definitive treatment of RCU by its author.
Jeff Preshing — locks and lock-free programming	Excellent on memory ordering in lock implementations. Pairs with CSCE 614 Module 12.

Module 05 — Deadlock

Resource	What it covers for this module
OSTEP — Chapter 32 (Common Concurrency Problems)	Deadlock and the non-deadlock bugs, with a study of what actually occurs in real code.
Linux kernel documentation — lockdep	How runtime lock-order validation works. Worth reading for the design even if you never use Linux.
Coffman, Elphick & Shoshani — System Deadlocks (1971)	The paper that identified the four conditions.
xv6 book — Chapter 6 (Locking), deadlock discussion	xv6's lock ordering and why it is documented where it is.

Module 06 — Memory Allocation

Resource	What it covers for this module
OSTEP — Chapter 17 (Free-Space Management)	Placement policies, coalescing, buddy and slab allocation, with worked examples.
Wilson et al. — Dynamic Storage Allocation: A Survey and Critical Review (free)	The definitive survey, and unusually sharp about how much allocator research measured the wrong things.
Bonwick — The Slab Allocator (free)	The original slab paper. Short, and the reasoning about object construction cost is still relevant.
mimalloc technical report (free)	A modern multithreaded allocator explained by its authors, including the remote-free problem.

Module 07 — Virtual Memory and Paging Policy

Resource	What it covers for this module
OSTEP — Chapters 21–24 (Beyond Physical Memory, Swapping, Policies)	Replacement policies with simulations, Belady's anomaly, and the working set model.

Resource	What it covers for this module
MIT 6.1810 — page tables lecture and the COW and lazy allocation labs	The labs are precisely Project 2's requirements.
xv6 book — Chapter 3 (Page tables)	The implementation, with the fault path through real code.
Denning — The Working Set Model for Program Behavior (free)	The original working set paper. Still the clearest statement of why locality makes paging work at all.

Module 08 — File Systems

Resource	What it covers for this module
OSTEP — Chapters 39–41 (Files and Directories, File System Implementation, FFS)	Inodes, directories, and allocation, with the Fast File System's locality arguments.
xv6 book — Chapter 8 (File system)	A complete, readable file system in about a thousand lines, organised in exactly the layers of §1.
MIT 6.1810 — file system lectures and lab	Including the lab that adds large files and symbolic links.
Linux Documentation — ext4 and VFS	How extents, delayed allocation, and the virtual file system layer work in production.

Module 09 — Crash Consistency

Resource	What it covers for this module
OSTEP — Chapter 42 (Crash Consistency: FSCCK and Journaling)	The clearest written treatment, including the three-write example and the journaling modes.
xv6 book — Chapter 8, logging section	A complete, working journaling layer in a few hundred lines. Read it before implementing your own.
Rosenblum & Ousterhout — The Design and Implementation of a Log-Structured File System (free)	The log-structured approach, which treats the whole disk as a log. Influential well beyond file systems.
Bonwick & Moore — ZFS: The Last Word in File Systems (free slides)	Copy-on-write, end-to-end checksums, and snapshots from the designers.

Module 10 — I/O and Device Drivers

Resource	What it covers for this module
OSTEP — Chapters 36–37 (I/O Devices, Hard Disk Drives)	Polling, interrupts, DMA, and disk scheduling with the seek-time analysis.

Resource	What it covers for this module
xv6 book — Chapter 5 (Interrupts and device drivers)	A complete disk and console driver, short enough to read entirely.
Linux Device Drivers, 3rd edition (free)	Dated in specifics and still the best free explanation of the concepts — memory-mapped I/O, DMA, deferred work.
Mogul & Ramakrishnan — Eliminating Receive Livelock (free)	The paper behind NAPI. Clear, and the problem it describes recurs wherever event rates grow.

Module 11 — Protection and Isolation

Resource	What it covers for this module
OSTEP — security chapters	Access control and the authentication model.
Linux kernel documentation — namespaces, cgroups v2, seccomp	The primary reference for what a container is actually built from.
Agache et al. — Firecracker: Lightweight Virtualization for Serverless Applications (free)	Why serverless platforms use micro-VMs rather than containers, argued with data.
Saltzer & Schroeder — The Protection of Information in Computer Systems (1975, free)	The paper that stated least privilege and the other design principles. Still the clearest statement of them.

Module 12 — Distributed Aspects: RPC and Network File Systems

Resource	What it covers for this module
OSTEP — Chapters 48–50 (Distributed Systems, NFS, AFS)	RPC, NFS's statelessness, and the AFS comparison, written clearly.
MIT 6.5840 Distributed Systems (free lectures and labs)	The natural follow-on. Build Raft and a fault-tolerant key-value store.
Waldo et al. — A Note on Distributed Computing (free)	The classic argument that local and remote objects cannot be unified. Short, and it settled the question.
Deutsch & Gosling — The Eight Fallacies of Distributed Computing	The original list, with commentary.

Module 13 — Kernel Design

Resource	What it covers for this module
Klein et al. — seL4: Formal Verification of an OS Kernel (free)	What was proved, how, and what it cost. The most significant result in kernel design in decades.

Resource	What it covers for this module
Liedtke — On micro-kernel construction (free)	The paper that showed microkernel IPC could be fast, and why the first generation was not.
The Tanenbaum–Torvalds debate (1992, archived)	The original argument, in full. Worth reading for how both sides were partly right.
Brendan Gregg — eBPF resources	What eBPF can actually do, from someone who uses it daily.
Madhavapeddy et al. — Unikernels: Library Operating Systems for the Cloud (free)	The case for collapsing the boundary entirely.